QMS Journal Club

Choong Pak Shen

February 22, 2026

QMS Journal Club 1st meeting: Linear algebra

Linear algebra

A vector $\vec{v}$ is an n-tuple of entries, which can be represented by a column vector,

$$\vec{v} = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}. \tag{1}$$

For our purposes, we focus only on complex vector space, which is the space of all complex vectors. The n-dimensional complex vector space is written as $\mathbb{C}^n$.

Vector space has two operations,

1. Vector addition, $\vec{v} = \vec{v}_1 + \vec{v}_2$
2. Scalar multiplication, $c\vec{v}$

A quantum bit (qubit) is a superposition (linear combination) of the basis states ($|0\rangle$, $|1\rangle$),

$$|\psi\rangle = \psi_0|0\rangle + \psi_1|1\rangle, \tag{2}$$

where ψ_0 and ψ_1 are called probability amplitudes, which are generally complex. The notation $|\psi\rangle$ is called a ket vector and can be represented as a column vector,

$$|\psi\rangle = \begin{pmatrix} \psi_0 \\ \psi_1 \end{pmatrix}, \tag{3}$$

where $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$. $|0\rangle$ and $|1\rangle$ are orthonormal to each other.

The dual to the ket vector is a bra vector,

$$\langle\psi| = \bar{\psi}_0\langle 0| + \bar{\psi}_1\langle 1| = \begin{pmatrix} \bar{\psi}_0 & \bar{\psi}_1 \end{pmatrix}, \tag{4}$$

where the overhead bar is complex conjugate operation.

The state vectors of a qubit live in the Hilbert space, which is a two-dimensional complex vector space $\mathcal{H} = \mathbb{C}^2$. The Hilbert space is equipped with an inner product operation,

$$\begin{aligned}\langle\psi|\psi\rangle &= \begin{pmatrix}\bar{\psi}_0 & \bar{\psi}_1\end{pmatrix}\begin{pmatrix}\psi_0 \\ \psi_1\end{pmatrix} \\ &= |\psi_0|^2 + |\psi_1|^2.\end{aligned} \tag{5}$$

There is also an outer product (tensor product),

$$\begin{aligned}|\psi\rangle\langle\psi| &= \begin{pmatrix}\psi_0 \\ \psi_1\end{pmatrix}\begin{pmatrix}\bar{\psi}_0 & \bar{\psi}_1\end{pmatrix} \\ &= \begin{pmatrix}|\psi_0|^2 & \psi_0\bar{\psi}_1 \\ \psi_1\bar{\psi}_0 & |\psi_1|^2\end{pmatrix}\end{aligned} \tag{6}$$

There exists four extremely important matrices, called the Pauli matrices:

$$\sigma_0 = I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \tag{7}$$

$$\sigma_1 = X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \tag{8}$$

$$\sigma_2 = Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \tag{9}$$

$$\sigma_3 = Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{10}$$

It is important to take note that

$$XY = iZ; YX = -iZ, \tag{11}$$

$$YZ = iX; ZY = -iX, \tag{12}$$

$$ZX = iY; XZ = -iY, \tag{13}$$

$$XX = YY = ZZ = I. \tag{14}$$

The Pauli matrices are traceless, i.e. the sum of diagonal elements is zero, $Tr(\sigma_i) = 0$.

It is also Hermitian, i.e. its conjugate transpose is equivalent to itself, $\sigma_i^\dagger = \sigma_i$, where $\dagger$ denotes conjugate transpose operation.

Let

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}; |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \tag{15}$$

$$|+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}; |-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \tag{16}$$

$$|+i\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}; |-i\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}. \tag{17}$$

The Pauli matrices have two eigenvalues ± 1,

$$X|+\rangle = |+\rangle ; X|-\rangle = -|-\rangle , \tag{18}$$
$$Y|+i\rangle = |+i\rangle ; Y|-i\rangle = -|-i\rangle , \tag{19}$$
$$Z|0\rangle = |0\rangle ; Z|1\rangle = -|1\rangle . \tag{20}$$

The Pauli matrices can be formulated by outer products,

$$X = |+\rangle\langle +| - |-\rangle\langle -| , \tag{21}$$
$$Y = |+i\rangle\langle +i| - |-i\rangle\langle -i| , \tag{22}$$
$$Z = |0\rangle\langle 0| - |1\rangle\langle 1| . \tag{23}$$

Previously, we mentioned that $\langle\psi|$ is the dual to $|\psi\rangle$. By using conjugate transpose operation, they are related by

$$\langle\psi| = (|\psi\rangle)^\dagger. \tag{24}$$

If A and B are two matrices, then $(AB)^\dagger = B^\dagger A^\dagger$. Therefore, $(A|\psi\rangle)^\dagger = \langle\psi|A^\dagger$.

Also, $(A + B)^\dagger = A^\dagger + B^\dagger$. Using this fact, one can easily show that the Pauli matrices are Hermitian, $\sigma_i^\dagger = \sigma_i$.

A matrix is unitary if $U^\dagger = U^{-1}$. The Pauli matrices are Hermitian and unitary.

Tensor product manifests itself in matrices as Kronecker product. Given two matrices A and B,

$$
\begin{aligned}
A \otimes B &= \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \otimes \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} \\
&= \begin{pmatrix} a_{11} \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} & a_{12} \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} \\ a_{21} \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} & a_{22} \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} \end{pmatrix} \\
&= \begin{pmatrix} a_{11}b_{11} & a_{11}b_{12} & a_{12}b_{11} & a_{12}b_{12} \\ a_{11}b_{21} & a_{11}b_{22} & a_{12}b_{21} & a_{12}b_{22} \\ a_{21}b_{11} & a_{21}b_{12} & a_{22}b_{11} & a_{22}b_{12} \\ a_{21}b_{21} & a_{21}b_{22} & a_{22}b_{21} & a_{22}b_{22} \end{pmatrix}. \qquad (25)
\end{aligned}
$$

In general, $A \otimes B \neq B \otimes A$.

If we have three vectors $|v_1\rangle, |v_2\rangle \in V$ and $|w\rangle \in W$,

$$(|v_1\rangle + |v_2\rangle) \otimes |w\rangle = |v_1\rangle \otimes |w\rangle + |v_2\rangle \otimes |w\rangle = |v_1 w\rangle + |v_2 w\rangle, \tag{26}$$

$$|w\rangle \otimes (|v_1\rangle + |v_2\rangle) = |w\rangle \otimes |v_1\rangle + |w\rangle \otimes |v_2\rangle = |wv_1\rangle + |wv_2\rangle. \tag{27}$$

If A acts on the vector space V and B acts on the vector space W, then

$$(A \otimes B)|v_1 w\rangle = A|v_1\rangle \otimes B|w\rangle. \tag{28}$$

Question: The Hadamard operator on one-qubit can be written as

$$H = \frac{1}{\sqrt{2}} \left[(|0\rangle + |1\rangle)\langle 0| + (|0\rangle - |1\rangle)\langle 1| \right]. \tag{29}$$

Show that the Hadamard transform on two qubits can be written as

$$H^{\otimes 2} = H \otimes H = \frac{1}{2} \sum_{x_1, x_2, y_1, y_2 \in \{0,1\}} (-1)^{x_1 y_1 + x_2 y_2} |x_1 x_2\rangle\langle y_1 y_2|. \tag{30}$$

$$
\begin{aligned}
H \otimes H = {} & \frac{1}{\sqrt{2}} \left[(|0\rangle + |1\rangle)\langle 0| + (|0\rangle - |1\rangle)\langle 1| \right] \otimes \\
& \frac{1}{\sqrt{2}} \left[(|0\rangle + |1\rangle)\langle 0| + (|0\rangle - |1\rangle)\langle 1| \right] \\
= {} & \frac{1}{2} [(|00\rangle + |01\rangle + |10\rangle + |11\rangle)\langle 00| \\
& + (|00\rangle - |01\rangle + |10\rangle - |11\rangle)\langle 01| \\
& + (|00\rangle + |01\rangle - |10\rangle - |11\rangle)\langle 10| \\
& + (|00\rangle - |01\rangle - |10\rangle + |11\rangle)\langle 11|] \\
= {} & \frac{1}{2} \sum_{x_1, x_2, y_1, y_2 \in \{0,1\}} (-1)^{x_1 y_1 + x_2 y_2} |x_1 x_2\rangle\langle y_1 y_2|.
\end{aligned}
$$

The trace of a matrix A is defined as the sum of its diagonal elements,

$$\begin{aligned} Tr(A) = Tr(IAI) &= Tr\left(\sum_{jk} |j\rangle\langle j|A|k\rangle\langle k|\right) \\ &= Tr\left(\sum_{jk} A_{jk}|j\rangle\langle k|\right) = \sum_{ijk} A_{jk}\langle i|j\rangle\langle k|i\rangle = \sum_{ijk} A_{jk}\langle k|i\rangle\langle i|j\rangle \\ &= \sum_{jk} A_{jk}\langle k|j\rangle = \sum_{jk} A_{jk}\delta_{kj} = \sum_{j} A_{jj}. \end{aligned} \tag{31}$$

The trace of a matrix satisfies a few properties:

1. $Tr(AB) = Tr(BA)$
2. $Tr(ABC) = Tr(BCA) = Tr(CAB)$
3. $Tr(A + B) = Tr(A) + Tr(B)$
4. $Tr(cA) = cTr(A)$
5. $Tr(A^T) = Tr(A)$
6. $Tr(A^T B) = Tr(AB^T) = Tr(B^T A) = Tr(BA^T)$
7. $Tr(A \otimes B) = Tr(A)Tr(B)$

Suppose $|\psi\rangle$ is a unit vector and A is an arbitrary operator.

$$\begin{aligned} Tr(A|\psi\rangle\langle\psi|) &= \sum_i \langle i|A|\psi\rangle\langle\psi|i\rangle \\ &= \sum_i \langle\psi|i\rangle\langle i|A|\psi\rangle \\ &= \langle\psi|A|\psi\rangle. \end{aligned} \tag{32}$$

QMS Journal Club 2nd meeting: Postulates of quantum mechanics and Bloch sphere I

Postulates of quantum mechanics

Postulate 1: Associated to any isolated physical system is a complex vector space with inner product, i.e. a Hilbert space, known as the state space of the system. The system is completely described by its state vector, which is a unit vector in the systems' state space.

Postulate 2: The evolution of a closed quantum system is described by a unitary transformation. That is, the state $|\psi\rangle$ of the system at time t_1 is related to the state $|\psi'\rangle$ of the system at time t_2 by a unitary operator U which depends only on the times t_1 and t_2,

$$|\psi'\rangle = U|\psi\rangle. \tag{33}$$

Explicitly, it can be described by the Schrodinger equation,

$$i\hbar\frac{d}{dt}|\psi\rangle = \hat{H}|\psi\rangle. \tag{34}$$

Or,

$$|\psi(t_2)\rangle = e^{\frac{-i\hat{H}(t_2-t_1)}{\hbar}}|\psi(t_1)\rangle. \tag{35}$$

Question: Suppose A and B are commuting Hermitian operators. Show that $e^A e^B = e^{(A+B)}$ by expanding the left and right hand sides until $n = 3$.

Question: Show that e^{iK} is unitary for some Hermitian matrix K. You only need to expand until $n = 6$.

Note that the matrix exponential is given by

$$e^A = \sum_{n=0}^{\infty} \frac{A^n}{n!}. \tag{36}$$

Postulate 3: Quantum measurements are described by a collection $\{M_m\}$ of measurement operators. These are operators acting on the state space of the system being measured. The index m refers to the measurement outcomes that may occur in the experiment. If the state of the quantum system is $|\psi\rangle$ immediately before the measurement then the probability that result m occurs is given by

$$p(m) = \langle\psi|M_m^\dagger M_m|\psi\rangle, \tag{37}$$

and the state of the system after the measurement is

$$\frac{M_m|\psi\rangle}{\sqrt{\langle\psi|M_m^\dagger M_m|\psi\rangle}}. \tag{38}$$

The measurement operators satisfies the completeness equation,

$$\sum_m M_m^\dagger M_m = I. \tag{39}$$

A projective measurement is described by an observable, M, a Hermitian operator on the state space of the system being observed. The observable can be written as

$$M = \sum_m m|m\rangle\langle m|, \tag{40}$$

where $P_m = |m\rangle\langle m|$ is the projector onto the eigenspace of M with eigenvalue m, satisfying $P_m P_{m'}^\dagger = \delta_{mm'} P_m$.
The expected value of the measurement is

$$\begin{aligned} E(M) = \langle M\rangle &= \sum_m m p(m) \\ &= \sum_m m\langle\psi|P_m|\psi\rangle \\ &= \langle\psi|\sum_m m|m\rangle\langle m||\psi\rangle \\ &= \langle\psi|M|\psi\rangle \end{aligned} \tag{41}$$

Question: The variance associated to observations of M, $[\Delta(M)]^2$ is given by

$$[\Delta(M)]^2 = \left\langle (M - \langle M \rangle)^2 \right\rangle . \tag{42}$$

Show that it is equivalent to $[\Delta(M)]^2 = \left\langle M^2 \right\rangle - \langle M \rangle^2$.

Question: Suppose A and B are two Hermitian operators, and $\langle\psi|AB|\psi\rangle = x + iy$. Show that $\langle\psi|[A,B]|\psi\rangle = 2iy$, $\langle\psi|\{A,B\}|\psi\rangle = 2x$.

The above implies that

$$|\langle\psi|[A,B]|\psi\rangle|^2 + |\langle\psi|\{A,B\}|\psi\rangle|^2 = 4|\langle\psi|AB|\psi\rangle|^2. \tag{43}$$

By the Cauchy-Schwarz inequality,

$$|\langle\psi|AB|\psi\rangle|^2 \leq \langle\psi|A^2|\psi\rangle\langle\psi|B^2|\psi\rangle. \tag{44}$$

Hence,

$$|\langle\psi|[A,B]|\psi\rangle|^2 \leq 4\langle\psi|A^2|\psi\rangle\langle\psi|B^2|\psi\rangle. \tag{45}$$

Question: Suppose C and D are two observables, and $A = C - \langle C \rangle$, $B = D - \langle D \rangle$. Obtain the Heisenberg's uncertainty principle,

$$\Delta(C)\Delta(D) \geq \frac{|\langle\psi|[C, D]|\psi\rangle|}{2}. \tag{46}$$

QMS Journal Club 3rd meeting: Postulates of quantum mechanics and Bloch sphere II

Phase

Global phase factor is a term $e^{i\theta}$ attached to $|\psi\rangle$ without changing the measurement statistics of $|\psi\rangle$,

$$\langle\psi|e^{-i\theta}M_m^\dagger M_m e^{i\theta}|\psi\rangle = \langle\psi|M_m^\dagger M_m|\psi\rangle. \tag{47}$$

For this reason, we say that $|\psi\rangle \sim e^{i\theta}|\psi\rangle$.

Relative phase is a term $e^{i\theta}$ attached between probability amplitudes,

$$|\psi\rangle = \alpha|\psi_\alpha\rangle + e^{i\theta}\beta|\psi_\beta\rangle. \tag{48}$$

Relative phase is a basis-dependent concept and may give rise to physically observable differences in measurement statistics.

For example, $|+\rangle = \frac{|0\rangle+|1\rangle}{\sqrt{2}}$ and $|-\rangle = \frac{|0\rangle-|1\rangle}{\sqrt{2}}$ are not the same up to a relative phase shift.

For one quantum bit (qubit), the Hilbert space is given as $\mathcal{H} = \mathbb{C}^2$, where

$$
\begin{aligned}
|\psi\rangle &= \psi_0|0\rangle + \psi_1|1\rangle, && (49)\\
&= \begin{pmatrix} \psi_0 \\ \psi_1 \end{pmatrix}, && (50)
\end{aligned}
$$

where ψ_0 and ψ_1 are complex numbers, and $|\psi_0|^2 + |\psi_1|^2 = 1$.

At first glance, there are 4 degrees of freedom. However, due to the normalization condition and the equivalence of global phase factor, we only need 2 degrees of freedom to parametrize a one-qubit state,

$$
|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle. \quad (51)
$$

Given the following Pauli matrices, find the expected values of Pauli-X, Y, and Z for $|\psi\rangle$.

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \; Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \; Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{52}$$

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle. \tag{53}$$

$$\langle\psi|X|\psi\rangle = \tag{54}$$
$$\langle\psi|Y|\psi\rangle = \tag{55}$$
$$\langle\psi|Z|\psi\rangle = \tag{56}$$

The expected values of Pauli matrices correspond to the spherical coordinates when the radius equals to 1,

$$\langle\psi|X|\psi\rangle = \sin\theta\cos\phi, \tag{57}$$

$$\langle\psi|Y|\psi\rangle = \sin\theta\sin\phi, \tag{58}$$

$$\langle\psi|Z|\psi\rangle = \cos\theta. \tag{59}$$

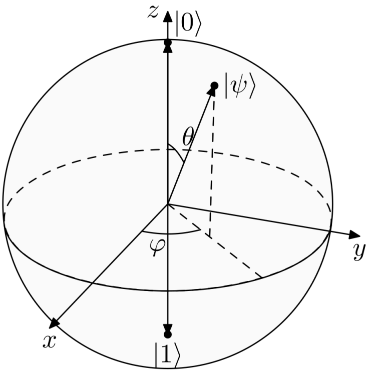


Figure 1: Bloch sphere

Postulates of quantum mechanics

Postulate 4: The state space of a composite physical system is the tensor product of the state spaces of the component physical systems. Moreover, if we have systems numbered 1 through n, and system number i is prepared in the state $|\psi_i\rangle$, then the joint state of the total system is

$$|\psi_1\rangle \otimes |\psi_2\rangle \otimes \ldots \otimes |\psi_n\rangle. \tag{60}$$

Postulate 4 enables us to define quantum entanglement. Suppose we are given two qubit states, $|\psi\rangle \in \mathcal{H}_A = \mathbb{C}^2$ and $|\phi\rangle \in \mathcal{H}_B = \mathbb{C}^2$. The two-qubit state $|\psi_{AB}\rangle$ is entangled if $|\psi_{AB}\rangle \neq |\psi\rangle \otimes |\phi\rangle$.
For example, the following state $|\psi_{AB}\rangle$ is separable,

$$\begin{aligned}
|\psi_{AB}\rangle &= \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle) \\
&= \frac{1}{2}[|0\rangle \otimes (|0\rangle + |1\rangle) + |1\rangle \otimes (|0\rangle + |1\rangle)] \\
&= \frac{1}{2}(|0\rangle + |1\rangle) \otimes (|0\rangle + |1\rangle) \\
&= |+\rangle \otimes |+\rangle
\end{aligned}$$

It is not always obvious to identify if a given two-qubit state is separable or not. One can derive separability condition for two qubits. As long as the two-qubit state satisfies the separability condition, the two-qubit state is separable (else, it is entangled).

Let $|\psi\rangle = \psi_0|0\rangle + \psi_1|1\rangle$, $|\phi\rangle = \phi_0|0\rangle + \phi_1|1\rangle$. Then

$$\begin{aligned} |\psi_{AB}\rangle &= |\psi\rangle \otimes |\phi\rangle \\ &= (\psi_0|0\rangle + \psi_1|1\rangle) \otimes (\phi_0|0\rangle + \phi_1|1\rangle) \\ &= \psi_0\phi_0|00\rangle + \psi_0\phi_1|01\rangle + \psi_1\phi_0|10\rangle + \psi_1\phi_1|11\rangle \end{aligned}$$

Notice that the probability amplitudes of $|00\rangle$ and $|11\rangle$, when multiplied together, is the same as the product of probability amplitudes of $|10\rangle$ and $|01\rangle$. We can conclude that for a two-qubit state $|\psi_{AB}\rangle = \psi_{00}|00\rangle + \psi_{01}|01\rangle + \psi_{10}|10\rangle + \psi_{11}|11\rangle$, if $\psi_{00}\psi_{11} = \psi_{10}\psi_{01}$, the two-qubit state is separable. Else, it is entangled.

Superdense coding

Can Alice communicate a number of classical bits of information by transmitting a smaller number of qubits, under the assumption that Alice and Bob receive an entangled resource beforehand?

- Alice and Bob initially share a pair of qubits in the Bell state,

$$|\psi\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}; \tag{61}$$

- If Alice wants to send 00, she does nothing; If Alice wants to send 01, she applies Z; If Alice wants to send 10, she applies X; If Alice wants to send 11, she applies iY;

- The final two-qubit state becomes:

$$00 : |\psi\rangle \to \frac{|00\rangle + |11\rangle}{\sqrt{2}} \tag{62}$$

$$01 : |\psi\rangle \to \frac{|00\rangle - |11\rangle}{\sqrt{2}} \tag{63}$$

$$10 : |\psi\rangle \to \frac{|10\rangle + |01\rangle}{\sqrt{2}} \tag{64}$$

$$11 : |\psi\rangle \to \frac{|01\rangle - |10\rangle}{\sqrt{2}} \tag{65}$$

- Alice sends her qubit to Bob;
- Bob performs a Bell basis measurement to determine which of the four possible bit strings Alice sent.

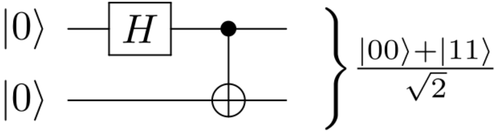


Figure 2: Bell state creation

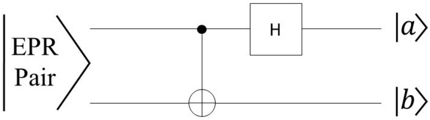


Figure 3: Bell basis measurement

QMS Journal Club 4th meeting: Pure and mixed states

Recap

1. A brief review on linear algebra
   - Basic operations on vectors and matrices
   - Pauli matrices
   - Inner product and outer product
   - Tensor product and Kronecker product
   - Trace of a matrix
2. Postulates of quantum mechanics
   - State space of a closed quantum system
   - Dynamics of a closed quantum system
   - Quantum measurement
   - Composite quantum system
   - Superdense coding

Consider a vertically polarized photon, given by

$$|V\rangle = \frac{|L\rangle + |R\rangle}{\sqrt{2}}. \tag{66}$$

When the photon passes through a circular polarizer that allows either only $|L\rangle$ or only $|R\rangle$ polarized light, half of the photons will be absorbed in both cases.

This may seem like half of the photons are in $|L\rangle$ or $|R\rangle$, but this is not true.

If the photon passes through a linear polarizer, there will be no absorption.

Meanwhile, if we pass unpolarized light through any type of polarizer, half of the intensity of light will be lost.

For the case of polarized light, the action of "measuring" the polarization of light is contextual, i.e. the outcome depends on what kind of measurement is being performed.

For the case of unpolarized light, the action of "measuring" the polarization of light is not contextual, because the outcome does not depend on the type of measurement being performed.

This suggests that for the case of polarized light, a quantum treatment is required (contextuality is a quantum phenomenon), whereas a classical treatment is required for the case of unpolarized light.

Polarized light is one example of pure states, whereby the probability of obtaining $|L\rangle$ or $|R\rangle$ is given by the absolute square of the probability amplitude.

Unpolarized light is one example of mixed states, and it is considered as a (classical) statistical ensemble, i.e. each photon having either $|R\rangle$ or $|L\rangle$ polarization with probability $\frac{1}{2}$.

For this example,

$$\rho = \frac{1}{2}|R\rangle\langle R| + \frac{1}{2}|L\rangle\langle L| = \frac{1}{2}|H\rangle\langle H| + \frac{1}{2}|V\rangle\langle V| = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \tag{67}$$

These two ensembles are completely indistinguishable experimentally, hence they are considered the same mixed state.

Conventionally,

$$|H\rangle = |0\rangle, \, |V\rangle = |1\rangle, \tag{68}$$
$$|D\rangle = |+\rangle, \, |A\rangle = |-\rangle, \tag{69}$$
$$|L\rangle = |+i\rangle, \, |R\rangle = |-i\rangle. \tag{70}$$

Formally, the density matrix of a quantum system is defined as

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|, \tag{71}$$

where $\sum_i p_i = 1$. We say that ρ is a mixture of pure states.

Density matrix is a more general formalism compared to state vector formalism, since a pure state $|\psi\rangle$ can be described by a density matrix $\rho = |\psi\rangle\langle\psi|$.

Consider

$$|\psi\rangle = \psi_0|0\rangle + \psi_1|1\rangle, \tag{72}$$

$$|\phi\rangle = \phi_0|0\rangle + \phi_1|1\rangle, \tag{73}$$

$$\rho = p_\psi|\psi\rangle\langle\psi| + p_\phi|\phi\rangle\langle\phi|, \tag{74}$$

where $p_\psi + p_\phi = 1$. Find ρ in a 2×2 matrix.

The density matrix is given by

$$\rho = \begin{pmatrix} p_\psi|\psi_0|^2 + p_\phi|\phi_0|^2 & p_\psi\psi_0\bar{\psi}_1 + p_\phi\phi_0\bar{\phi}_1 \\ p_\psi\bar{\psi}_0\psi_1 + p_\phi\bar{\phi}_0\phi_1 & p_\psi|\psi_1|^2 + p_\phi|\phi_1|^2 \end{pmatrix}. \tag{75}$$

If we compute $\langle 0|\rho|0\rangle$, we get $p_\psi|\psi_0|^2 + p_\phi|\phi_0|^2$.

The diagonal element gives the probability of finding a particle in the corresponding basis state $\{|0\rangle, |1\rangle\}$.

For a general state $|\Psi\rangle$,

$$\begin{aligned} \langle\Psi|\rho|\Psi\rangle &= p_\psi\langle\Psi|\psi\rangle\langle\psi|\Psi\rangle + p_\phi\langle\Psi|\phi\rangle\langle\phi|\Psi\rangle \\ &= p_\psi|\langle\psi|\Psi\rangle|^2 + p_\phi|\langle\phi|\Psi\rangle|^2 \end{aligned} \tag{76}$$

it can be seen that $\langle\Psi|\rho|\Psi\rangle$ gives the total probability of finding a particle in the pure state $|\Psi\rangle$ within a mixture ρ.

A density matrix ρ has the following properties:

1. $Tr(\rho) = 1$;
2. $\rho^\dagger = \rho$;
3. It is a positive semi-definite matrix, i.e. $\langle\psi|\rho|\psi\rangle \geq 0$.

For pure states, the density matrix ρ further satisfies the following:

1. $\rho^2 = \rho$;
2. $Tr(\rho^2) = 1$.

How many real parameters do you need to describe a 2×2 density matrix?

You need three real parameters to describe a 2×2 density matrix.

Consider again the one-qubit state,

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle. \tag{77}$$

Its density matrix is given by

$$\rho = \begin{pmatrix} \cos^2\left(\frac{\theta}{2}\right) & e^{-i\phi}\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right) \\ e^{i\phi}\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right) & \sin^2\left(\frac{\theta}{2}\right) \end{pmatrix} \tag{78}$$

Previously, we learn that

$$P_x = \langle\psi|X|\psi\rangle = \sin\theta\cos\phi, \tag{79}$$

$$P_y = \langle\psi|Y|\psi\rangle = \sin\theta\sin\phi, \tag{80}$$

$$P_z = \langle\psi|Z|\psi\rangle = \cos\theta. \tag{81}$$

We want to rewrite the density matrix in terms of $\{P_x, P_y, P_z\}$, which are called the polarization (Bloch) vectors,

$$\begin{aligned}
\rho &= \begin{pmatrix} \frac{\cos\theta+1}{2} & \frac{\sin\theta(\cos\phi - i\sin\phi)}{2} \\ \frac{\sin\theta(\cos\phi + i\sin\phi)}{2} & \frac{1-\cos\theta}{2} \end{pmatrix} \\
&= \frac{1}{2}\begin{pmatrix} P_z + 1 & P_x - iP_y \\ P_x + iP_y & 1 - P_z \end{pmatrix} \\
&= \frac{1}{2}\left(I + P_x X + P_y Y + P_z Z\right) \\
&= \frac{1}{2}\left(I + \vec{P}\cdot\vec{\sigma}\right),
\end{aligned} \tag{82}$$

where

$$\vec{P} = \begin{pmatrix} P_x \\ P_y \\ P_z \end{pmatrix}, \vec{\sigma} = \begin{pmatrix} X \\ Y \\ Z \end{pmatrix}. \tag{83}$$

In general, a density matrix can be written as $\rho = \frac{1}{2}\left(I + \vec{r} \cdot \vec{\sigma}\right)$.

Density matrix formalism is useful when

1. the preparation of a system can randomly produce different pure states, and thus one must deal with the statistics of the ensemble of possible preparations;
2. when one wants to describe a physical system that is entangled with another, without describing their combined state (i.e. finding the reduced density matrix of a subsystem).

QMS Journal Club 5th meeting: Density matrix formalism and reduced density matrices

Previously, we mentioned that density matrix formalism is useful when

1. the preparation of a system can randomly produce different pure states, and thus one must deal with the statistics of the ensemble of possible preparations;
2. when one wants to describe a physical system that is entangled with another, without describing their combined state (i.e. finding the reduced density matrix of a subsystem).

Consider the Bell state

$$|\psi\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}. \tag{84}$$

The description of this Bell state is that when measurement is performed for qubit A, there is a probability $\frac{1}{2}$ to get $|0\rangle$ or $|1\rangle$.

How can we describe the quantum state for qubit A alone?

One might be tempted to write

$$|\psi_A\rangle = \frac{1}{\sqrt{2}}(|0_A\rangle + |1_A\rangle). \tag{85}$$

This description gives you the correct statistics when you measure in the Pauli Z basis, but it gives you a wrong statistics when you measure in the Pauli X basis because $|\psi_A\rangle$ is now its eigenstate, $|+\rangle$.

This again shows that the state vector formalism does not completely capture the classical statistical ensemble aspect of mixed states.

Previously, an example of maximally mixed state was given,

$$\rho = \frac{1}{2}|R\rangle\langle R| + \frac{1}{2}|L\rangle\langle L| = \frac{1}{2}|H\rangle\langle H| + \frac{1}{2}|V\rangle\langle V| = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \tag{86}$$

Regardless of what measurement basis we are taking, we always get a probability of $\frac{1}{2}$ to get either $\{|R\rangle, |L\rangle\}$ (or $\{|H\rangle, |V\rangle\}$).

Postulates of quantum mechanics

Postulate 1: Associate to any isolated physical system is a complex vector space with inner product, known as the state space of the system. The system is completely described by its density operator, which is a positive semi-definite operator $\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$ with trace one, acting on the state space of the system.

Postulate 2: The evolution of a closed quantum system is described by a unitary transformation. The state ρ of the system at time t_1 is related to the state ρ' of the system at time t_2 by a unitary operator U,

$$\rho' = U\rho U^\dagger. \tag{87}$$

Postulate 3: Quantum measurements are described by a collection of measurement operators M_m, such that the probability that result m occurs for the state of a quantum system ρ is given by

$$p(m) = Tr(M_m^\dagger M_m \rho), \tag{88}$$

and the state of the system after the measurement is

$$\frac{M_m \rho M_m^\dagger}{Tr(M_m^\dagger M_m \rho)}. \tag{89}$$

For projective measurement $M = \sum_m m|m\rangle\langle m|$ with projector $P_m = |m\rangle\langle m|$, $P_m^\dagger P_m = P_m$ and hence

$$\begin{aligned} p(m) &= \sum_i p_i \langle\psi_i|P_m|\psi_i\rangle = \sum_i p_i \langle\psi_i|P_m \sum_j |j\rangle\langle j|\psi_i\rangle \\ &= \sum_{ij} p_i \langle j|\psi_i\rangle\langle\psi_i|P_m|j\rangle = Tr\left(\sum_i p_i |\psi_i\rangle\langle\psi_i|P_m\right) \\ &= Tr(\rho P_m) = Tr(P_m \rho). \end{aligned} \tag{90}$$

The expectation value of a measurement M follows the same argument and can be written as

$$\langle M \rangle = Tr(M\rho). \tag{91}$$

Postulate 4: The state space of a composite quantum system is the tensor product of the state spaces of the component quantum systems, such that the joint state of the total system is $\rho_1 \otimes \rho_2 \otimes \ldots \otimes \rho_n$.
Suppose that a density matrix is given as

$$\rho = \frac{3}{4}|0\rangle\langle 0| + \frac{1}{4}|1\rangle\langle 1|. \tag{92}$$

Let

$$|a\rangle = \sqrt{\frac{3}{4}}|0\rangle + \sqrt{\frac{1}{4}}|1\rangle, \tag{93}$$

$$|b\rangle = \sqrt{\frac{3}{4}}|0\rangle - \sqrt{\frac{1}{4}}|1\rangle. \tag{94}$$

Then,

$$\rho = \frac{3}{4}|0\rangle\langle 0| + \frac{1}{4}|1\rangle\langle 1| = \frac{1}{2}|a\rangle\langle a| + \frac{1}{2}|b\rangle\langle b|. \tag{95}$$

Different ensembles of quantum states can give rise to the same density matrix. When will two sets of state vectors $|\psi\rangle$ and $|\phi\rangle$ generate the same density matrix?

Theorem: The sets $|\psi_i\rangle$ and $|\phi_j\rangle$ generate the same density matrix if and only if

$$|\psi_i\rangle = \sum_j U_{ij}|\phi_j\rangle, \tag{96}$$

where U_{ij} is a unitary matrix.

Reduced density matrix

Suppose we have a quantum system A and B, defined by ρ_{AB}. The reduced density operator for system A is defined by

$$\rho_A = Tr_B(\rho_{AB}), \tag{97}$$

where Tr_B is called partial trace over system B. Partial trace is defined by

$$Tr_B(|a_1\rangle\langle a_2| \otimes |b_1\rangle\langle b_2|) = |a_1\rangle\langle a_2| Tr(|b_1\rangle\langle b_2|) = |a_1\rangle\langle a_2|\langle b_2|b_1\rangle. \tag{98}$$

Question 1: Consider the Bell state $|\psi_{AB}\rangle = \frac{|00\rangle+|11\rangle}{\sqrt{2}}$. Find the reduced density matrix of the second qubit.

Question 2: Suppose a composite quantum system A and B is in the state $|a\rangle|b\rangle$. Show that the reduced density matrix of system A is a pure state.

Quantum teleportation

Quantum teleportation begins with two parties, Alice and Bob, sharing a pair of entangled qubits in one of the Bell bases,

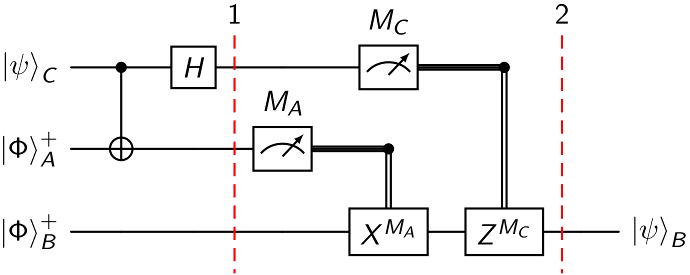
$$|\Phi\rangle_{AB}^{+} = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle), \tag{99}$$

$$|\Phi\rangle_{AB}^{-} = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle), \tag{100}$$

$$|\Psi\rangle_{AB}^{+} = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle), \tag{101}$$

$$|\Psi\rangle_{AB}^{-} = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle). \tag{102}$$

Without loss of generality, we focus on the Bell basis $|\Phi\rangle_{AB}^{+}$ in our discussion. The other Bell bases follow the same mathematical arguments.


1
2
M_C
M_A
$|\psi\rangle_C$
$|\Phi\rangle_A^+$
$|\Phi\rangle_B^+$
H
X^{M_A}
Z^{M_C}
$|\psi\rangle_B$

Initially, the shared Bell state between Alice and Bob, and the qubit that Alice wants to send to Bob is given as:

$$\begin{aligned} |\psi\rangle_{CAB} &= (\alpha\,|0\rangle_C + \beta\,|1\rangle_C) \otimes \frac{1}{\sqrt{2}}(|00\rangle_{AB} + |11\rangle_{AB}) \\ &= \frac{1}{\sqrt{2}}(\alpha\,|000\rangle + \alpha\,|011\rangle + \beta\,|100\rangle + \beta\,|111\rangle). \end{aligned} \tag{103}$$

After the CNOT gate, the above quantum state becomes:

$$|\psi\rangle_{CAB} = \frac{1}{\sqrt{2}}(\alpha\,|000\rangle + \alpha\,|011\rangle + \beta\,|110\rangle + \beta\,|101\rangle). \tag{104}$$

After the Hadamard gate, the quantum state at Step 1 will become:

$$
\begin{aligned}
|\psi\rangle_{CAB} &= \frac{1}{\sqrt{2}}\left[\frac{\alpha}{\sqrt{2}}(|0\rangle_C + |1\rangle_C) \otimes |00\rangle_{AB} + \frac{\alpha}{\sqrt{2}}(|0\rangle_C + |1\rangle_C) \otimes |11\rangle_{AB}\right. \\
&\qquad \left. + \frac{\beta}{\sqrt{2}}(|0\rangle_C - |1\rangle_C) \otimes |10\rangle_{AB} + \frac{\beta}{\sqrt{2}}(|0\rangle_C - |1\rangle_C) \otimes |01\rangle_{AB}\right] \\
&= \frac{1}{2}[\alpha\,|000\rangle + \alpha\,|100\rangle + \alpha\,|011\rangle + \alpha\,|111\rangle + \beta\,|010\rangle - \beta\,|110\rangle \\
&\quad + \beta\,|001\rangle - \beta\,|101\rangle] \\
&= \frac{1}{2}[|00\rangle_{CA}\,(\alpha\,|0\rangle_B + \beta\,|1\rangle_B) + |01\rangle_{CA}\,(\beta\,|0\rangle_B + \alpha\,|1\rangle_B) \\
&\quad + |10\rangle_{CA}\,(\alpha\,|0\rangle_B - \beta\,|1\rangle_B) + |11\rangle_{CA}\,(-\beta\,|0\rangle_B + \alpha\,|1\rangle_B)].
\end{aligned}
\tag{105}
$$

In Step 2, quantum measurements will be performed on Alice's qubits (qubit A and C) and the appropriate Pauli correction will be applied to Bob's qubit (qubit B) to recover the quantum information that is teleported. In short,

1. If the quantum measurement results in $|00\rangle_{CA}$, nothing ($\sigma_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$) will be performed;
2. If the quantum measurement results in $|01\rangle_{CA}$, Pauli X, $\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ will be performed;
3. If the quantum measurement results in $|10\rangle_{CA}$, Pauli Z, $\sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ will be performed;
4. If the quantum measurement results in $|11\rangle_{CA}$, Pauli Y up to a global phase of i, $i\sigma_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ will be performed.

Find the density matrix of $|\psi\rangle_{CAB}$,

$$|\psi\rangle_{CAB} = \frac{1}{2}[|00\rangle (\alpha|0\rangle + \beta|1\rangle) + |01\rangle (\beta|0\rangle + \alpha|1\rangle) + |10\rangle (\alpha|0\rangle - \beta|1\rangle) + |11\rangle (-\beta|0\rangle + \alpha|1\rangle)]. \tag{106}$$

Verify that when Alice's system is being traced out, the reduced density matrix of Bob is maximally mixed.

This shows that any measurements performed by Bob will contain no information about $|\psi\rangle_{CAB}$, and Alice cannot use quantum teleportation to transmit information to Bob faster than speed of light.

Here are some intermediate steps for your verifications. Let

$$
\begin{aligned}
|B_1\rangle &= \alpha|0\rangle + \beta|1\rangle, \\
|B_2\rangle &= \beta|0\rangle + \alpha|1\rangle, \\
|B_3\rangle &= \alpha|0\rangle - \beta|1\rangle, \\
|B_4\rangle &= -\beta|0\rangle + \alpha|1\rangle.
\end{aligned}
$$

Then

$$
\begin{aligned}
\rho_{AB} = \frac{1}{4}[&|0\rangle\langle 0| \otimes |B_1\rangle\langle B_1| + |0\rangle\langle 1| \otimes |B_1\rangle\langle B_2| + |1\rangle\langle 0| \otimes |B_2\rangle\langle B_1| \\
&+ |1\rangle\langle 1| \otimes |B_2\rangle\langle B_2| + |0\rangle\langle 0| \otimes |B_3\rangle\langle B_3| + |0\rangle\langle 1| \otimes |B_3\rangle\langle B_4| \\
&+ |1\rangle\langle 0| \otimes |B_4\rangle\langle B_3| + |1\rangle\langle 1| \otimes |B_4\rangle\langle B_4|]. \qquad (107)
\end{aligned}
$$

Lastly,

$$
\begin{aligned}
\rho_B &= \frac{1}{4}[|B_1\rangle\langle B_1| + |B_2\rangle\langle B_2| + |B_3\rangle\langle B_3| + |B_4\rangle\langle B_4|] \\
&= \frac{1}{4}[2(|\alpha|^2 + |\beta|^2)|0\rangle\langle 0| + 2(|\alpha|^2 + |\beta|^2)|1\rangle\langle 1|] = \frac{1}{2}I. \qquad (108)
\end{aligned}
$$

QMS Journal Club 6th meeting: Schmidt decomposition, bipartite entanglement, and state purification

Higher order tensors

From one-qubit state

$$|\psi_A\rangle = \psi_0|0\rangle + \psi_1|1\rangle \tag{109}$$

to two-qubit state

$$|\psi_{AB}\rangle = \psi_{00}|00\rangle + \psi_{01}|01\rangle + \psi_{10}|10\rangle + \psi_{11}|11\rangle, \tag{110}$$

the probability amplitude of the quantum states changes from an one-index object to a two-index object.
In general, one-index object $\Psi_A = (\psi_i)$ is called tensor of order 1, while two-index object $\Psi_{AB} = (\psi_{ij})$ is called tensor of order 2.
Tensor of order 1 is usually represented as a column vector and it coincides with the state vector representation.
Meanwhile, tensor of order 2 is represented as a matrix. In two-qubit case,

$$\Psi_{AB} = (\psi_{ij}) = \begin{pmatrix} \psi_{00} & \psi_{01} \\ \psi_{10} & \psi_{11} \end{pmatrix}. \tag{111}$$

Schmidt decomposition

Theorem: Suppose $|\psi\rangle$ is a pure state of a composite system AB. Then there exist orthonormal states $|i_A\rangle$ for system A, and orthonormal states $|i_B\rangle$ for system B, such that

$$|\psi\rangle = \sum_i \lambda_i |i_A\rangle |i_B\rangle, \tag{112}$$

where λ_i are non-negative real numbers satisfying $\sum_i \lambda_i^2 = 1$, known as Schmidt coefficients.

Schmidt decomposition is a consequence of singular value decomposition when representing $|\psi\rangle$ as a tensor of order 2,

$$\Psi_{AB} = U \Lambda V^T, \tag{113}$$

where U and V are unitary matrices and Λ is a diagonal matrix.

Singular value decomposition works for rectangular matrices. As a consequence of this, Schmidt decomposition works for any bipartite quantum systems of different dimension, for example qubit-qutrit system.

As a convention, we usually require an ordering condition, i.e. $\lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_n$. This makes sure that the unitary matrices U and V to be uniquely determined.

For two qubits, the reduced density matrices ρ_A and ρ_B are

$$\rho_A = \begin{pmatrix} |\psi_{00}|^2 + |\psi_{01}|^2 & \psi_{00}\bar{\psi}_{10} + \psi_{01}\bar{\psi}_{11} \\ \bar{\psi}_{00}\psi_{10} + \bar{\psi}_{01}\psi_{11} & |\psi_{10}|^2 + |\psi_{11}|^2 \end{pmatrix}, \tag{114}$$

$$\rho_B = \begin{pmatrix} |\psi_{00}|^2 + |\psi_{10}|^2 & \psi_{00}\bar{\psi}_{01} + \psi_{10}\bar{\psi}_{11} \\ \bar{\psi}_{00}\psi_{01} + \bar{\psi}_{10}\psi_{11} & |\psi_{01}|^2 + |\psi_{11}|^2 \end{pmatrix}. \tag{115}$$

Question: Show that

$$\rho_A = \Psi_{AB}\Psi_{AB}^\dagger, \tag{116}$$

$$\rho_B = \Psi_{AB}^T\bar{\Psi}_{AB}. \tag{117}$$

As a result, when singular value decomposition is applied,

$$\rho_A = \Psi_{AB}\Psi_{AB}^\dagger = U\Lambda V^T \bar{V}\Lambda U^\dagger = U\Lambda^2 U^\dagger, \tag{118}$$

$$\rho_B = \Psi_{AB}^T\bar{\Psi}_{AB} = V\Lambda U^T \bar{U}\Lambda V^\dagger = V\Lambda^2 V^\dagger. \tag{119}$$

This shows that U and V are the eigendecomposition of the reduced density matrices ρ_A and ρ_B respectively, and the reduced density matrices ρ_A and ρ_B have the same eigenvalues (singular values or Schmidt coefficients).

For two qubits, there are two Schmidt coefficients λ_1 and λ_2, and we know that $\lambda_1^2 + \lambda_2^2 = 1$. Therefore, there are only three possibilities,

1. Maximally entangled states: $\lambda_1^2 = \lambda_2^2 = \frac{1}{2}$
2. Entangled states: $\lambda_1^2 > \lambda_2^2 \neq 0$
3. Separable states: $\lambda_1^2 = 1$, $\lambda_2^2 = 0$

Question: Classify the following two-qubit states according to the framework.

1. $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$
2. $\frac{1}{\sqrt{3}}(|00\rangle + |01\rangle + |10\rangle)$
3. $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$

One can also describe the separability of two-qubit states by finding the separability condition. Let $|\psi_A\rangle \in \mathcal{H}_A$ and $|\phi_B\rangle \in \mathcal{H}_B$, and

$$|\psi_A\rangle = \psi_0|0\rangle + \psi_1|1\rangle, \tag{120}$$
$$|\phi_B\rangle = \phi_0|0\rangle + \phi_1|1\rangle. \tag{121}$$

The combined two-qubit state becomes

$$\begin{aligned} |\psi_{AB}\rangle &= |\psi_A\rangle \otimes |\phi_B\rangle \\ &= \psi_0\phi_0|00\rangle + \psi_0\phi_1|01\rangle + \psi_1\phi_0|10\rangle + \psi_1\phi_1|11\rangle. \end{aligned} \tag{122}$$

One can notice that the product of probability amplitudes of $|00\rangle$ and $|11\rangle$ is the same as the product of probability amplitudes of $|01\rangle$ and $|10\rangle$, when the two-qubit state is separable.

Therefore, the separability condition of two qubits is given by

$$\psi_{00}\psi_{11} - \psi_{01}\psi_{10}. \tag{123}$$

If $\psi_{00}\psi_{11} - \psi_{01}\psi_{10} = 0$, the two-qubit state is separable. Otherwise, it is entangled.

Last but not least, notice that the separability condition of two qubits is the same as the determinant of Ψ_{AB},

$$\Psi_{AB} = (\psi_{ij}) = \begin{pmatrix} \psi_{00} & \psi_{01} \\ \psi_{10} & \psi_{11} \end{pmatrix}. \tag{124}$$

The whole process is called entanglement classification of two qubits, which is well-studied for bipartite systems due to Schmidt decomposition.

From group theoretic perspective, the action of U and V is called local unitary group action on two qubits. Notice that U only acts on qubit A, V only acts on qubit B. Hence, singular value decomposition (or equivalently, Schmidt decomposition) on a two-qubit state can be written as

$$\begin{aligned}
U \otimes V|\psi_{AB}\rangle &= \sum_{ij} U \otimes V \psi_{ij}|ij\rangle \\
&= \sum_{ijklmn} \psi_{ij}|k\rangle\langle k|U|m\rangle\langle m| \otimes |l\rangle\langle l|V|n\rangle\langle n|(|i\rangle \otimes |j\rangle) \\
&= \sum_{ijklmn} \psi_{ij}|k\rangle\langle k|U|m\rangle\langle m|i\rangle \otimes |l\rangle\langle l|V|n\rangle\langle n|j\rangle \\
&= \sum_{ijkl} \psi_{ij}|k\rangle\langle k|U|i\rangle \otimes |l\rangle\langle l|V|j\rangle \\
&= \sum_{ijkl} \psi_{ij} u_{ki} v_{lj}|k\rangle \otimes |l\rangle = \sum_{ijkl} u_{ki}\psi_{ij} v^T_{jl}|k\rangle \otimes |l\rangle \qquad (125)
\end{aligned}$$

Question: How many real parameters do you need to write down a 2×2 unitary matrix of determinant 1?

You need 3 real parameters to define a 2×2 unitary matrix,

$$U = \begin{pmatrix} u_{11} & u_{12} \\ -\bar{u}_{12} & \bar{u}_{11} \end{pmatrix}. \tag{126}$$

In order to specify a generic two-qubit state, you need 6 real parameters (ignoring the global phase).

Hence, it is possible for the local unitary group of two qubits to decompose a generic two-qubit state into the Schmidt coefficients, which are real and non-negative.

Question: Can we have Schmidt decomposition for three qubits?

State purification

Suppose we are given a mixed state ρ_A. It is possible to introduce another (reference) system, R, in such a way that the combined quantum state $|AR\rangle$ is a pure state, and $\rho_A = Tr_R(|AR\rangle\langle AR|)$.
Due to Schmidt decomposition, ρ_A can be written as

$$\rho_A = \sum_i p_i |i_A\rangle\langle i_A|. \tag{127}$$

The combined system needs to be in this form

$$|AR\rangle = \sum_i \sqrt{p_i}|i_A\rangle|i_R\rangle. \tag{128}$$

The partial trace Tr_R of the combined system is given by

$$\begin{aligned} Tr_R(|AR\rangle\langle AR|) &= \sum_{ij} \sqrt{p_i p_j}|i_A\rangle\langle j_A| \, Tr(|i_R\rangle\langle j_R|) = \sum_{ij} \sqrt{p_i p_j}|i_A\rangle\langle j_A|\delta_{ij} \\ &= \sum_i p_i |i_A\rangle\langle i_A| = \rho_A \end{aligned} \tag{129}$$